

AKM Sabbir

DATA SCIENCE · PROJECT MANAGEMENT · DATA ANALYTICS · DATA MANAGEMENT

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About Me

- I love to work with data engineering, data analytics, data and project management
- I work to evaluating and assessing risk, profit and potential benefit using models, statistics and analytical tools
- I have been sharing my experiences through blogging, classes, presentation, talks, publications
- Managed projects and teams using agile philosophy and Scrum Framework
- I worked on high performance computing, data centric engineering & analytics
- I worked on web api development and cloud based service development

Professional History

ML team in Infolytx Limited

Dhaka, BD

SENIOR SOFTWARE ENGINEER II, PROJECT MANAGEMENT

June. 2022 - April. 2024

We have been working on product that helps to assign medical disease code on text written by professional doctors, project distribution to teams, skilled people identification from national ID information

Responsibilities

- R&D on disease code identification from doctors notes, gathered data and did initial feasibility study
- Developed APIs post, get, put methods for single request & batch processing using multi process & async functions
- helped in data annotation to build a dataset and designed rule based, knowledge based text labeling algorithm

Achievement datasets, rule based algorithm, service apis for text annotations, large scale data processing, front end UI, fine-tuning LLM for large scale models

Natural language processing E-Gov. project in Reve Systems

Dhaka, BD

PROJECT MANAGER

Nov. 2019- May 2022

Developing large Bangla language corpus, spell and grammar checking service

Responsibilities

- Collected data from news papers, law farms, education institutions, BD administration to build large Bangla corpus
- Designed a probabilistic, contextual Bangla spell checker
- Designed Bangla punctuation, space and conjunction fixing algorithm

Achievement Designed Bangla tokenizer, stemmer, context based spell checker, punctuation validation, missing space identification between words for Bangla language.

Dispatch Efficiencies, Verizon Wireless

Spring Valley, NY U.S.A

MACHINE LEARNING ENGINEER

Nov. 2018 - Dec 2019

We were developing a ML based automatic troubleshooting and recovery process for FIOS services in Verizon nation-wide optical network architecture

Responsibilities

- Developing a data processing system that will accelerate processed and structured data delivery to algorithm
- Developing reusable library and platform for sharing with our other teams

Achievement A Efficient library for data access, process and delivery to algorithm team.

Data Quality, NPD Group

Port Washington NY U.S.A

MACHINE LEARNING ENGINEER

May. 2017 - Oct. 2018

We are designing and developing a distributive data processing and information extraction system

Responsibilities

- Developed a data processing engine for a distributive map reduce process using spark framework
- Developed a generative text classification algorithm using variation of TF-IDF and Bayesian model
- Developed an optimized search algorithm to facilitates the searching from 2.7 Billions of Items, processed 4TB data
- Designed own dependency and build resolution tool

Achievement Our application process the data, classify correct brand, model and category using entire population

Samsung Research and Development Institute

Dhaka, Bangladesh

SOFTWARE ENGINEER

Aug. 2011 - Aug. 2013

Worked with the Samsung innovation team to do research on embedded system and find out new ways of application

Responsibilities

- Novel feature extension of Samsung applications
- Developed Region specific solution such as text rendering, bi-directionality, application UI layout correction
- Code quality and format checking

Achievement we ensured the software quality was always met with full compliance to Samsung's requirements

GrameenPhone Ltd(Telenor)

Dhaka, Bangladesh

SYSTEM ENGINEER

Aug. 2008 - Aug. 2010

Worked with the data quality team. Flawless data collection, verification and storage using internal infrastructure

Responsibilities

- Collected data from thousands of regional office hosting mysql servers and merge them to store in central repository
- Worked with C# socket programming, TCP protocol, ICMP protocol for network access control

Achievement Uninterrupted data collection of daily sales and archiving without any corruption, ensured data quality

Publications

- [1] AKM Sabbir, Sayed Fahmid Islam, Md. Monir; Effect of Inflation Due to Rising Energy Price on Consumer Purchasing Behavior in Bangladesh. On going research under BUP.
- [2] AKM Sabbir;Why Generate Data From Noisy Distribution in Generative AI; <https://medium.com/GenAI>
- [3] AKM Sabbir; GAN : How To Learn with Generative Adversial Network; <https://medium.com/GAN>
- [4] AKM Sabbir;Asynchronous Non-Blocking I/O to Achieve Single Thread Concurrency;<https://medium.com/AsynchronousInputOutput>
- [5] AKM Sabbir;Dr. Antonio Jimeno Yepes; Dr. Ramakanth Kavuluru; Knowledge Based Word Sense Disambiguation using Neural Word and Concept Embeddings. 2017 IEEE 17th International Conference on Bioinformatics and Bio-engineering(BIBE).
- [6] Dr. Ramakanth Kavuluru, AKM Sabbir, Toward Automated E-cigarette Surveillance: Spotting E-cigarette Proposals on Twitter 2016, Journal Proceeding at JBI(Journal of Biomedical Informatics) Manuscript. No: JBI-15-563R2. (media coverage of our work)
- [7] AKM Sabbir, Dr. Sally Ellingson, Side-Effect Term Matching for Computational Adverse Drug Reaction Predictions. Proceedings of the 15th Annual UT-KBRIN Bioinformatics Summit 2016: Cadiz, KY, USA. 8-10 April 2016.
- [8] AKM Sabbir, Khondoker Asif Akter, A Novel Door Detection Technique using graph cut, expectation maximization, surf Feature, Support Vector Machine and set unification. WECS 2013, BUET, Dhaka, Bangladesh

Skills

Programming Language	Python, JavaScript, JAVA, C/C++, Scala, Sql, mySql, LaTeX
Web Back End	ExpressJS, Spring boot, Fast Api Python, J2EE Servlet,
Web Front End	React JS, Bootstrap, CSS
Embedded Frameworks	Android, FreeRTOS
Big Data Preprocessing	Hive, Spark, Hadoop, Hbase, Dynamo DB, mpi4py, Elastic Search, Oracle
Machine Learning Framework	Tensorflow, Pytorch, Keras, scikit-learn, spark, LLM, LangChain, HayStack
Version Control	Git, GitLab, Bitbucket
Linux Scripting	Bash Shell
Cloud Service	AWS, EC2, DynamoDB, S3, IAM, RDS
DevOps	Dockers, Bitbucket pipeline, Jenkins
Project Management	Jira

Education

Bangladesh University of Professionals

Dhaka

MASTER OF BUSINESS ADMINISTRATION (EXECUTIVE-MBA) - BUSINESS ANALYTICS

Jan. 2024 -

- Majoring in Organizational management, Strategic policy making and Business analytics

University of Illinois

Online Urbana Champaign

SPECIALIZATION IN BUSINESS VALUE AND PROJECT MANAGEMENT

Feb. 2023 - Feb. 2024

- Majoring in organizational structure, governance, strategic planning, ethics and principles in capitalism

University of Kentucky

Lexington KY USA

M.S. IN COMPUTER SCIENCE

Aug. 2013 - Dec. 2016

- Awarded Research Assistant-ship from Bio statistics Dept.
- Worked as research assistant with Merkey Cancer Center under college of Medicine Dept.

Khulna University of Engineering and Technology

Khulna Bangladesh

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Mar. 2004 - March 2008

- Ranked among the top five students and got the Engineering and Technical Scholarship

Research Experience

College of Medicine, University of Kentucky

RESEARCH ASSISTANT AT MERKEY CANCER CENTER

Lexington KY U.S.A

Jan. 2016 - May 2016

Worked on data centric drug discovery algorithms.

Responsibilities

- Conducted research on drug efficacy and developed side effect prediction model
- Developed text similarity matching algorithm using semantic and syntactic analysis

Achievement We were able to develop a list of drug protein pair having similar side effects, did poster presentation

Department of Bio-statistics at University of Kentucky

RESEARCH ASSISTANT

Lexington, KY U.S.A

Aug. 2013 - May. 2015

We worked on community detection from the social network data, word sense disambiguation etc

Responsibilities

- Built dataset for classification, label assignment and text mining using resources UMLS, PubMed, MEDLINE, Twitter
- Used Deep Neural Network such as Convolutional Neural Net to perform text classification
- Developed and published novel predictive model that infer decisions for biomedical and social network data.

Achievement We published couple of peer reviewed journal and conference paper

Department of Computer Science UK

TEACHING ASSISTANT

Lexington KY, U.S.A

Aug. 2016 - Dec. 2016

Conducted matlab simulation lab

Responsibilities

- Delivered Lecture and conducted class at lab
- Prepared materials for simulation lab classes

Achievement we helped student to develop a background on effective utilization of programming.

Honors & Awards

INTERNATIONAL

2019	My Research work has been cited in 75 different conference and journal paper (google_citation),	New York, USA
2007	Graduate Research Assistant at UK(USA)(Fall, Spring, Summer),	Lexington, U.S.A

DOMESTIC

2004	Khulna University of Eng & Tech Technical Scholarship 04, 05, 06 & 07 (tuition and 500 \$ year),	Khulna, Bangladesh
2006	Runner up Southern Regional Programming Contest (SRPC),	Khulna Univ. Bangladesh
2006	ACM Int'l Collegiate Programming Contest, Regional Contest, Placed among top ten teams	Dhaka, Bangladesh
2007	ACM Int'l Collegiate Programming Contest, Regional Contest, Ranked among top ten	Dhaka, Bangladesh